NADH

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Identity Sheet

Property	Value
Name	NADH (Reduced Nicotinamide Adenine Dinucleotide)
Other names	DPNH, Coenzyme I (reduced form), NAD+ (oxidised form)
Chemical formula	$C_{21}H_{29}N_7O_{14}P_2$
Molecular weight	665.44 g/mol
CAS number	58-68-4
PubChem CID	928
Appearance	White to yellow powder; solutions are colourless (NAD+) or fluorescent (NADH)
Solubility	Freely soluble in water

Structural formula:

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Nicotinamide–Ribose–PO4–PO4–Ribose–Adenine
(reduced nicotinamide ring carries extra H at C4)
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NADH is the reduced form of NAD+, the central electron carrier in cellular metabolism. NADH is **intrinsically fluorescent** (Ex 340 nm / Em 460 nm), while NAD+ is not, making NADH a natural endogenous fluorophore. In the urinary context, NADH fluorescence has been explored as a marker of cellular metabolic activity and as a component of the overall urine autofluorescence spectrum.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
NAD+	$C_{21}H_{27}N_7O_{14}P_2$	663.43	Oxidised form; NOT fluorescent at 340/460 nm
NADPH	$C_{21}H_{30}N_7O_{17}P_3$	745.42	Phosphorylated form; similar fluorescence; anabolic role
FAD	$C_{27}H_{33}N_9O_{15}P_2$	785.55	Flavin-based coenzyme; fluorescent at 450/525 nm (different wavelength)
Riboflavin (B2)	$C_{17}H_{20}N_4O_6$	376.36	FAD precursor; strong fluorescence at 450/525 nm

Medical Information

Origin

Endogenous: Cellular Metabolism NADH is produced in all cells via glycolysis (cytoplasm), pyruvate dehydrogenase, and the TCA cycle (mitochondria). It is the primary electron donor to Complex I of the electron transport chain. Intracellular NADH concentrations: ~100-400 uM. NADH is predominantly intracellular; extracellular/urinary levels reflect cell lysis, metabolic overflow, and renal tubular cell turnover.

Exogenous NADH supplements are available commercially. Dietary NAD+ precursors include niacin (vitamin B3) and nicotinamide riboside (NR), but these do not directly contribute urinary NADH.

Primary & Secondary Biological Roles

Primary role: - **Electron carrier in catabolism:** NADH donates electrons to the mitochondrial ETC for ATP production (2.5 ATP per NADH).

Secondary roles: - **Redox signalling:** NAD+/NADH ratio regulates sirtuins, PARPs, and metabolic gene expression. - **Autofluorescence biomarker:** NADH fluorescence is used in tissue metabolic imaging (optical biopsy).

Catabolism and Elimination Pathway

- NADH is rapidly oxidised to NAD+ in mitochondria (ETC) or cytoplasm (lactate dehydrogenase).
- Free NADH is unstable in acidic conditions (t1/2 ~30 min at pH 4).
- Urinary NADH derives from cellular debris, lysed cells, and renal tubular cell turnover.
- NAD+ and its metabolites (nicotinamide, NMN, ADPR) are also excreted.

Expression in Humans

Normal Levels

Compartment	Reference Range
Intracellular NADH	100-400 uM
Blood NAD+ total	~20-40 uM
Urine NADH	Trace (not routinely measured); detectable by fluorescence

Factors Influencing Levels **Increased urinary NADH fluorescence:** - Increased cellular turnover or damage - UTI (bacterial metabolic activity) - Cancer cells (altered metabolism, Warburg effect) - Mitochondrial disorders

Decreased: - Not clinically significant in isolation.

Associated Pathologies NADH itself is not a standalone clinical biomarker. Its fluorescence contributes to urinary autofluorescence patterns studied in research contexts for:

Condition	NADH Relevance	Notes
Bladder cancer screening	Altered NADH/FAD ratio in exfoliated cells	Research stage
UTI	Bacterial NADH fluorescence	Experimental
Metabolic disorders	Altered redox state	Research stage

Presence in Urine

Should it be normally present? In **trace amounts** – from normal cellular turnover. Not routinely measured.

Normal urinary levels: Not standardised. Detectable as fluorescence signal at Ex 340 / Em 460 nm.

Form in urine: Free NADH molecule (unstable, degrades in acidic urine).

Solubility: Freely soluble.

Detection in Urine

Available Clinical Assays

No routine clinical assays measure urinary NADH. It is a research biomarker detected by autofluorescence.

Optimal Urine Type for Measurement

- Fresh specimen required (NADH is unstable). Analyse immediately.
- Protect from light and heat.
- Neutral or slightly alkaline pH preserves NADH stability.

Actual Gold Standard

No established clinical gold standard. **HPLC-UV at 340 nm** or **HPLC-fluorescence (Ex 340 / Em 460 nm)** are research reference methods.

Interferences in Measurement

Interference	Effect	Mechanism
Acidic pH	Degradation	NADH decomposes at pH <6
Light	Photodegradation	UV-sensitive
Other fluorophores	Spectral overlap	Riboflavin, FAD, proteins fluoresce at nearby wavelengths
Oxidation	Signal loss	NADH -> NAD+ (non-fluorescent)

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV absorption:** NADH absorbs at **340 nm** (epsilon ~6,220 M-1cm-1). NAD+ absorbs at 260 nm but not 340 nm. The 340 nm band is diagnostic for the reduced form.
  - LOD: ~1 uM by direct UV.
  - Sample prep: None (direct measurement of fresh urine).

Fluorescence Detection This is the primary detection modality for NADH.

- **Intrinsic fluorescence: Ex 340 nm / Em 460 nm** (blue fluorescence). Quantum yield ~2%.
  - LOD: ~0.1-1 uM in buffer; ~5-10 uM in urine (matrix quenching).
  - Lifetime: ~0.4 ns (free NADH), ~1-3 ns (protein-bound NADH).
- **Two-photon excitation:** Ex 720-740 nm (two-photon) / Em 460 nm. Used in tissue imaging.
- **FLIM (Fluorescence Lifetime Imaging):** Distinguishes free vs bound NADH by lifetime. Not applicable for bulk urine.
- **Sample prep:** Fresh, pH-neutral urine; protect from light.

Raman Detection

Peak (cm-1)	Assignment
~1030	Nicotinamide ring C-H stretch
~1340	Adenine ring
~1580	Nicotinamide ring C=C/C=N
~1690	C=O stretch (amide)

- Conventional Raman: LOD ~1 mM (too high for urinary levels).
- SERS: Potentially LOD ~1-10 uM. Few studies on urinary NADH specifically.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1690	Amide C=O	Nicotinamide
~1580	Ring C=C	–
~1080	PO4 stretch	Phosphodiester

- Not practical for urinary NADH (too low concentration, overlapping bands).

Voltammetry Detection

- NADH is electrochemically oxidised: NADH -> NAD+ + H+ + 2e-
- **Oxidation potential:** +0.5-0.7 V vs Ag/AgCl on bare carbon; reduced to +0.1-0.3 V on modified electrodes (Meldola Blue, carbon nanotubes, graphene).
- LOD: ~0.1-1 uM on nanostructured electrodes.
- **Interference:** Ascorbic acid, uric acid oxidise at similar potentials.

Other Detection Technologies

1. **Enzymatic cycling assays:** NAD+/NADH quantified by lactate dehydrogenase cycling with colourimetric or fluorimetric endpoint. LOD: ~10 nM. Measures total NAD pool.
2. **LC-MS/MS:** Definitive identification and quantification. LOD: ~1 nM. Sample prep: protein precipitation, extraction.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
UV absorption	~1 uM	340 nm	None	Direct, simple
Fluorescence	~0.1-1 uM	Ex 340/Em 460 nm	None (fresh)	Primary method
Electrochemical	~0.1-1 uM	+0.1-0.3 V (modified)	Buffer	Real-time
Enzymatic cycling	~10 nM	Colorimetric/fluorimetric	Enzyme kit	Ultra-sensitive
LC-MS/MS	~1 nM	m/z 664->514	Extraction	Definitive
Raman/FTIR	>1 mM	–	–	Not practical

References

1. PMC - Evaluating Cell Metabolism Through Autofluorescence of NAD(P)H and FAD. https://ncbi.nlm.nih.gov/pmc/articles/PMC6352511/
2. PMC - NADH and NADPH Autofluorescence. https://ncbi.nlm.nih.gov/pmc/articles/PMC5145803/
3. Wiley - NADH Autofluorescence as Bioenergetic Marker. https://onlinelibrary.wiley.com/doi/full/10.1002/cyto.a.23597
4. PubChem - NADH, CID 928. https://pubchem.ncbi.nlm.nih.gov/compound/928